

The Sphere Is an Orbit

Dimension as Group Action, and the Price of a Metric

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Abstract

A sphere is not an object situated in space. It is the orbit of a single point under the rotation group, and every property conventionally attributed to it follows from that fact alone. This paper takes the orbit as primary and derives the rest, then prices what a metric costs when it is not assumed.

Three consequences are immediate. An orbit has no privileged element, so a sphere supplies no coordinate and no address until a mark is made. The centre is the unique fixed point of the group action, so every direction there lies in a single orbit, no direction is distinguishable from any other, and no gradient is definable without first breaking the symmetry that defines the centre. And dimension is orbit dimension: the $(n-1)$ -sphere is what the rotation group of n -space does to one point, so the familiar progression from line to circle to sphere is one generator applied repeatedly rather than three objects assembled in order of complexity.

Quotienting three-space by the rotation group leaves the half-line. All rotation-respecting structure in three dimensions is therefore a function on a ray, which is the exact sense in which a nested family of spheres carries a single datum, and the fixed point is that ray's endpoint.

The second half prices the metric. Distinguishable states with lawful transitions between them constitute a directed graph and nothing more. Minimum directed path length on that graph satisfies identity, positivity, and the triangle inequality — three axioms forced by concatenation — and fails symmetry generically, in 64.6% of reachable pairs measured here. What these primitives generate is a quasi-metric, which is precisely a category enriched over the non-negative reals under addition, and symmetry is not among the enrichment axioms. It is not a missing theorem: a symmetric distance quotients direction, and direction is what distinguishes transformation from stasis.

Symmetry is bought with reversibility. A scale is bought with a mark. The continuum is bought with completeness, and unbounded addressing with irrationality. The integers come free from concatenation and the rationals free from comparison; nothing above them does. Riemannian geometry is accordingly not primitive but the reversible, scaled, complete sector of a directed transformation structure, and the symmetric sphere is a special case rather than a starting point.

Contents

1. Shape first, value second	4
1.1 The chain reads in both directions.....	4
1.2 What the ordering forbids.....	5
2. The sphere is an orbit	5
2.1 An orbit has no privileged element	5
2.2 The centre is the unique fixed point.....	6
2.3 Dimension is orbit dimension.....	7
3. What follows for the interior.....	7
3.1 The orbit space is a ray.....	8
3.2 The two readings of one boundary.....	8
4. The cost of a mark.....	9
5. Contact, phase, and the price of a turn	10
5.1 Contact is how a mark arrives	10
5.2 Three motions at a contact, and only one is phase.....	11
5.3 The free direction is the untransmittable direction	12
5.4 Abstract and material: the same change, two prices.....	12
6. What distinction and continuation generate	13
6.1 Four axioms are forced	13
6.2 Symmetry is not forced, and fails generically.....	13
6.3 The result is a known object.....	14
6.4 Symmetry is forbidden, not merely underived.....	14
7. The price list	15
7.1 The integers and the rationals are free	15
7.2 The reals cost two separate things.....	15
7.3 What completeness is for.....	16
8. What is recoverable without an exterior	16
8.1 Two independent constructions, one invariant	17
8.2 A weaker starting point still suffices	17
8.3 The criterion: invariance decides.....	18
9. Matter, and the two kinds of state.....	19
10. Consequences	19
10.1 Riemannian geometry is a sector, not a foundation	19

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10.2 Space is a reading, not a container.....	20
10.3 Magnitude begins as a count.....	20
11. Relation to existing work	20
11.1 What is not claimed.....	21
11.2 What is claimed.....	21
12. Claims and their status	22
13. What must follow	23
14. Summary.....	24

1. Shape first, value second

The ordering used throughout this paper is the reverse of the conventional one, and stating it before anything else prevents the argument from silently reverting to habit.

shape -> available relations -> address space -> operators -> values

Read conventionally, a geometric object sits inside a space that is already equipped with coordinates, a metric, and a full apparatus of operations, and the object's properties are read off using machinery that was in place before the object arrived. Under that reading, mathematics is ambient. It fills the space; objects are placed into it; the objects have values because the ambient machinery assigns them.

The claim here is that the dependency runs the other way. A shape does not contain mathematics and does not have values waiting to be extracted. It determines which distinctions can be made without violating its own constraints, and the mathematics available at that shape is exactly the set of operations those distinctions support. Values are the last thing to appear in the chain, not the first.

This is not a philosophical preference and it can be checked. The equation $x^2 + y^2 + z^2 = r^2$ is not what a sphere is. Writing it presupposes three coordinate axes, an origin, a squaring operation, and a metric, and a sphere supplies none of these. The equation describes a sphere that has already been placed inside a structure the sphere did not provide. Section 2 gives the object that requires no such placement.

1.1 The chain reads in both directions

The same relation traversed from opposite ends produces two different-looking accounts, and conflating them is the most common way the ordering gets lost.

Direction	Chain	What appears fundamental
Abstract \u2192 material	logic \u2192 relation \u2192 mathematical structure \u2192 shape \u2192 manifestation	the relation
Material \u2192 abstract	matter \u2192 shape \u2192 admissible relations \u2192 recoverable mathematics	the shape

Logic is invariant and everywhere. Mathematics is not a substance distributed through space; it is the abstract description of relational structure. Shape is what that structure looks like once it is constrained enough to be manifest. Reading from the abstract end, geometry appears derived and relation appears primary. Reading from the material end, geometry appears primary because it is the first thing encountered. Both readings are of one relation, and the disagreement about which is fundamental is an artefact of where the reader started.

This matters for the rest of the paper because the two halves work in opposite directions. Sections 2 through 4 begin with a group action and derive what a sphere supplies — the abstract-to-material reading. Sections 5 through 7 begin with distinguishable states and derive what a metric costs — the material-to-abstract reading. They meet at the same place: a structure that supplies fewer distinctions than the usual apparatus assumes, and a list of what the missing distinctions cost.

1.2 What the ordering forbids

Three moves are unavailable under this ordering and are not used anywhere below.

- Placing an object into a pre-existing coordinate system and reading properties from that system. The coordinate system is downstream of the object's own constraints, not a container for them.
- Treating a value as a property the object carries. A value is what an operation returns, and the operation must first be admissible at that shape.
- Deriving a shape from an equation. The equation is what becomes writable once the shape has supplied the distinctions its symbols refer to.

2. The sphere is an orbit

Take one point and one group. Apply every element of the rotation group to that point and collect the results.

$$S = \{ g \cdot p : g \text{ in } SO(3) \}$$

The set obtained is the sphere through p . Nothing was assembled. No points were placed, no surface was swept out by a construction, and no ambient three-dimensional container was required for the definition. A single point was moved, and the sphere is the complete record of where it can go.

This is the standard definition of an orbit, and the claim of this section is that it is the entire content of the word sphere. Everything conventionally said about spheres — that they have no corners, that every point is like every other, that they are determined by one number, that they enclose a volume — is either an immediate consequence of the orbit structure or, in the case of enclosure, a statement about a different object entirely.

The economy of this is worth pausing on. The conventional definition of a sphere is the set of points at fixed distance from a centre, which requires a notion of distance, a designated centre, and an ambient space in which both make sense. The orbit definition requires a group and a point. The group is the transformation structure; the point is what it acts on. Distance does not appear, and Section 7 shows it can be recovered afterward rather than being needed in advance.

2.1 An orbit has no privileged element

The rotation group acts transitively on the orbit: given any two of its points, some rotation carries the first to the second. The construction is explicit and worth writing out, because it is what makes the claim checkable rather than merely asserted.

Let u and v be unit vectors. Set $a = u \times v$, $s = |a|$, and $c = u \cdot v$. If s is nonzero, form the skew matrix K with entries built from the components of a , and set

$$R = I + K + K^2 * (1 - c) / s^2$$

This is Rodrigues' rotation formula for the rotation about the axis a carrying u onto v . It is orthogonal with determinant one, so it lies in $SO(3)$, and it satisfies $Ru = v$ by construction. The degenerate cases are $s = 0$, where u and v are parallel and either the identity or a half-turn about any perpendicular axis suffices.

Applied to 500 randomly sampled pairs on the orbit, the construction produced a valid rotation carrying the first point to the second in every case, with the resulting matrix orthogonal to within 10^{-9} . Transitivity is not

an empirical finding here — the formula proves it — but the check confirms the implementation and rules out an error in the reading.

Consequence: no point of a sphere is distinguishable from any other by any property the group respects.

There is no origin, no first point, no canonical zero, and no basis for calling one location special. This is not a difficulty in locating a distinguished point; it is the statement that the object contains none to locate. Every coordinate system ever placed on a sphere begins by choosing an element of a set that has no distinguished element, and that choice is a mark whose cost Section 4 computes exactly.

The same argument explains a familiar fact usually attributed to a defect in the coordinate system. Latitude and longitude fail at the poles — longitude becomes undefined there — and this is routinely described as an unfortunate artefact of a particular chart. It is not. The sphere is not homeomorphic to the plane, so no chart covers it without singularity, and the failure is the object declining to supply something a global flat coordinate system would require.

2.2 The centre is the unique fixed point

The origin is fixed by every rotation. This is immediate: rotations are linear and a linear map sends the zero vector to itself. Verified over 2000 random group elements as a check on implementation, and it holds identically.

No other point is fixed by every rotation. Suppose q is nonzero and fixed by all of $SO(3)$. Then in particular it is fixed by every rotation about an axis perpendicular to q , and such a rotation by any nonzero angle moves q . So q cannot be fixed, and the origin is the unique fixed point.

The consequence for structure at that point is the part that matters. A gradient is a distinguished direction carrying a magnitude. At the fixed point, the group acts on the directions and does so transitively — every direction can be carried to every other by some rotation that leaves the point itself in place. So no direction at the fixed point is distinguishable from any other by any property the group respects, and a gradient cannot be defined there without first breaking the symmetry that defines the point.

At the fixed point no gradient is definable. This is group theory, not physics: a gradient requires a distinguished direction and the group makes every direction equivalent.

The consequence for weight is a statement about the domain of definition of an operation, and it should be stated carefully because the word invites a physical reading it does not carry. Weight is the response to a gradient. Where no gradient is definable, the weighting operation has no argument, so the fixed point admits no weighting relation. This is not the same as carrying zero weight: zero is a value the operation returns, and this is a case where the operation does not apply. The distinction is the one between a function evaluating to zero and a function being undefined outside its domain.

Newtonian gravity is untouched by any of this and supplies its own, stronger result in the same place. The shell theorem gives a field of exactly zero everywhere inside a uniform spherical shell, not merely at the centre. A direct numerical evaluation over a shell of 200,000 uniformly distributed mass elements gives field magnitudes at the level of sampling noise at radii 0.0, 0.2, 0.5, 0.9, and 0.99, while immediately outside at radius 1.01 the field is 1.068 against the predicted $1/R^2 = 0.980$, and at radius 10 it is 9.998×10^{-3} against 10^{-2} .

Gauss's law explains why the interior is silent and does so in exactly the vocabulary of this paper. The flux of the field through any closed surface is fixed by the mass enclosed. The field at a point is therefore defined by a surface integral rather than by the volume surrounding it, and inside a hollow shell the enclosed mass is zero, so the flux is zero, so the field is zero. The interior contributes nothing except through what it encloses, and this is the divergence theorem rather than an approximation.

2.3 Dimension is orbit dimension

The construction generalises without modification. The orbit of a point under the rotation group of n -space is the $(n-1)$ -sphere.

Group	Acting on	Orbit of one point	Orbit dimension	Stabiliser
$SO(2)$	$\mathbb{R}^2$	S^1 , the circle	1	trivial
$SO(3)$	$\mathbb{R}^3$	S^2 , the sphere	2	$SO(2)$
$SO(4)$	$\mathbb{R}^4$	S^3	3	$SO(3)$
$SO(n)$	$\mathbb{R}^n$	S^{n-1}	$n-1$	$SO(n-1)$

The dimension count follows from the orbit-stabiliser relation. The orbit of a point is the group modulo the subgroup fixing that point, so its dimension is $\dim SO(n) - \dim SO(n-1)$. Since $SO(n)$ has dimension $n(n-1)/2$, the difference is $n-1$, which is the dimension of the sphere. Nothing about an embedding is used.

So the familiar progression is not three objects stacked in order of complexity:

line $\rightarrow$ rotate $\rightarrow$ circle $\rightarrow$ rotate $\rightarrow$ sphere

It is one generator applied twice. The circle is not something added to the line and the sphere is not something added to the circle. Each is the same relation read at another turn, and what changes between them is the orientation of the reading rather than the count of objects in play. Nothing is constructed and nothing travels into a new space. The reader changes orientation and calls the resulting face a new object.

Dimension, on this account, counts the orbit rather than the container. The habit of saying that a sphere is a two-dimensional object living in three-dimensional space inverts the dependency: the two-dimensionality is the orbit's own, computed from the group, and the three-dimensional ambient space is a convenience for drawing it.

3. What follows for the interior

The orbit is a set of points at fixed radius. The interior is not in it.

Points at different radii lie in different orbits, and the radius is the label distinguishing one orbit from another. So the interior of a sphere is not a region that the sphere contains. It is a family of other orbits, with the fixed point as the degenerate member consisting of a single element. What is habitually called the inside is a stack of concentric orbits terminating at a fixed point, and the sphere itself is one member of that stack rather than a container for the rest.

This is why the interior supplies no relation. Within a single orbit every point is equivalent to every other under the group, so nothing internal distinguishes them. Between orbits the only difference is the label. There is no third kind of structure available.

3.1 The orbit space is a ray

That the label is a single number is forced rather than observed, and the derivation is short. The orbit space is the quotient of the ambient space by the group action — the set of orbits, with two points identified whenever some group element carries one to the other.

$$\mathbb{R}^3 / SO(3) = [0, \infty)$$

The map sending a point to its distance from the origin is constant on orbits, since rotations preserve length, and it separates distinct orbits, since two points of equal length lie on a common sphere and are therefore connected by a rotation. So the quotient map is a bijection onto the non-negative reals, and the orbit space is the half-line.

The whole of three-space, read through rotation, collapses to a ray. Every structure in three dimensions that respects the rotation group — every rotationally invariant function, field, or measure — is a function on that ray and nothing more. This is the exact sense in which the nested family carries one datum, and it explains why a rotationally symmetric problem in three variables reduces to one variable: the reduction is not a technique but a statement about the quotient.

The endpoint of the ray is the fixed point. It is the only orbit with a single element, the only place where the group acts trivially, and the boundary of the quotient. Everything said in Section 2.2 about the centre is a statement about that endpoint.

3.2 The two readings of one boundary

A closed surface embedded in three-space divides the complement into exactly two connected components, one bounded and one unbounded, with the surface as their common boundary. This is the Jordan-Brouwer separation theorem, it holds for any closed embedded surface however convoluted, and it requires no mark, no coordinate system, and no choice.

Three features of this deserve separation because they are habitually collapsed. The count is exactly two, always. Nothing was added to obtain it — the surface was not cut, marked, or measured, and closure alone forces the bifurcation. And the two components are not symmetric: one has finite extent, the other does not.

There is a second asymmetry superimposed on the first and it is a different kind of thing. At the zero-thickness boundary the inward-facing and outward-facing surfaces are the same point set. What distinguishes them is the orientation of the normal, and reversing the reader from interior to exterior replaces n with $-n$. Bounded against unbounded is not that. No reversal of sign converts a finite region into an infinite one. The two asymmetries coexist on the same boundary at different orders, and conflating them costs the argument its precision.

A boundary is one object with two readings, not two places with a wall between them.

This is the central structural claim of the paper. Write the closure as a pair of relational states rather than as a surface separating two regions:

$$B = (B \text{ read from within } , B \text{ read from without })$$

The two entries are not two surfaces and not two spaces. They are the same closure under complementary orientation. What is outside under one reading is inside under the other, and neither reading has priority, because the orientation of a normal is a choice made by a reader and not a property carried by the surface.

The consequence is that the nesting does not terminate by reaching a smallest sphere and does not require an ambient region beyond the largest. It folds. The outermost resolved boundary is the interior state of the next reading, and there is no absolute outside in the sense of a further region awaiting description. The apparent outside of a closure is the complementary state of that same closure, and asking what lies beyond it is asking for a third reading of an object that has two.

This is the point at which the paper's ordering does real work. Under the conventional reading, a sphere is an object in space and its outside is more of the same space, so the question of what lies beyond the largest sphere is well posed and unanswered. Under the orbit reading the question is malformed: there is no ambient space to be beyond, because the sphere was never in one.

4. The cost of a mark

Marks consume symmetry. The symmetry is finite-dimensional, so the ledger closes exactly, and the closing point can be computed rather than estimated.

The computation is a rank calculation. A rotation infinitesimally close to the identity acts on a point m by adding $w \times m$, where w is an element of the Lie algebra — a three-dimensional space. Requiring a set of marks to be fixed means requiring $w \times m = 0$ for every mark m , which is a linear system in w . The dimension of the surviving stabiliser is three minus the rank of that system.

Marks	Rank of the system	Continuous stabiliser	What becomes available
0	0	$SO(3)$, dimension 3	the radius, and nothing else
1	2	$SO(2)$, dimension 1	an axis; the equator is still one orbit
2 generic	3	trivial, dimension 0	an angle; a great-circle plane
2 antipodal	2	$SO(2)$, dimension 1	nothing the first mark did not give
3 off-plane	3	trivial, mirror removed	chirality; the frame is fixed
4 and beyond	3	already trivial	content within a frame, never more frame

Three rows in that table are forced and are worth stating as claims rather than as entries.

The first mark leaves a full circle of rotational freedom. It establishes an axis, not an angle. Every point around that axis remains in a single orbit of the residual group and therefore remains indistinguishable from every other. This is the geometric content of a statement that recurs across the framework: one distinction gives an origin and leaves the rotation free.

The second mark does not select an angle. It creates the angle. Before it there is no angle to select, because all 360 degrees lie in one orbit and orbits have no distinguished elements. A reader who imagines the second mark as picking a value from a range has smuggled in the range.

The third mark resolves a discrete ambiguity rather than a continuous one, and this is why the ledger has a remainder after the continuous dimensions are spent. Two marks leave the reflection through the plane containing them and the centre — an improper orthogonal transformation, determinant -1 , fixing both marks. A third mark off that plane removes it. So the full accounting is three continuous dimensions plus one discrete bit, and the marks spend two, then one, then the bit.

The antipodal case is the sharp edge and it shows that the count of marks is not the quantity being spent. Two marks at opposite poles leave the stabiliser at dimension one, not zero, because they lie on the same axis and the second supplies nothing the first did not. What is consumed is symmetry, and a mark placed on structure that already exists is free and produces nothing. This is the geometric form of a principle that holds throughout: a distinction that does not distinguish is not a distinction.

Saturation is real and it is at three. A fourth mark adds no frame. It adds content within a frame that already exists, which is a different kind of information and does not enlarge the coordinate structure. So a body's capacity to carry a coordinate system is finite, exactly countable, and equal to the dimension of the symmetry group its shape possesses.

5. Contact, phase, and the price of a turn

Section 4 counted what a mark costs without saying how a mark arrives. It arrives by contact, and following that through connects two results that otherwise sit in the paper unrelated: the minimality of spherical contact and the residual symmetry left by a first mark turn out to be the same fact.

5.1 Contact is how a mark arrives

Compute the dimension of the contact set for pairs of convex bodies and the sphere is distinguished again, for the same reason it was distinguished in Section 2.

Contact	Dimension of the contact set
Sphere against sphere	0 \u2014 a point
Sphere against plane	0 \u2014 a point
Cylinder against plane	1 \u2014 a line
Cube against cube, corner to face	0 \u2014 a point, but non-generically
Cube against cube, edge to face	1 \u2014 a line
Cube against cube, face to face	2 \u2014 a square

The sphere is the only convex body whose contact with any other convex body is generically zero-dimensional. A cube can meet another at a point, but only in the non-generic corner configuration; perturb it and the contact becomes a line or a face. A sphere cannot be perturbed into a higher-dimensional contact because it has no flat region to lie along and no straight edge to lie on.

The same property produces this and the isoperimetric minimality: having no flat region is why the boundary is minimal for the enclosed volume and why any contact is a single point. Minimum boundary and minimum aperture are one property read at two scales.

And a point of contact is a mark. It is a distinguished location on each surface, established by relation to something the surface is not. So two spheres in external tangency each acquire exactly one mark, and by the ledger of Section 4 each drops from $SO(3)$ to $SO(2)$. The pair retains one continuous freedom: relative rotation about the axis through both centres. Verified — the tangency point and both centres are fixed under every rotation about that axis.

One contact does not fix a frame. It fixes an axis and leaves a circle free.

This is Section 4's first-mark result arrived at from the dynamics side rather than the geometric one. The reader is not standing outside the ledger observing it. The reader is the ledger, spending symmetry one contact at a time, and the count is the same three.

5.2 Three motions at a contact, and only one is phase

Two spheres in tangency, each rotating, admit exactly three independent relative motions, and separating them shows which one carries the free circle.

Let the common normal at the contact be n , and let the two bodies have angular velocities ω_1 and ω_2 . Write the relative angular velocity as $\omega = \omega_1 - \omega_2$ and decompose it against the normal. The component along n is spin; the component perpendicular to n is roll. Separately, the surface velocities at the contact point may fail to match, and that mismatch is slip.

Configuration	Spin	Roll (tangential)	Slip at the contact point
Both bodies still	0	0	0
Equal spin about the normal	0	0	0
Opposite spin \u2014 twist	2	0	0
Counter-roll \u2014 rolling, no slip	0	2 (about x)	0
Same roll \u2014 slipping	0	0	2 (about y)

The third row is the one that matters. Under pure twist, the roll is zero and the slip is zero, and the contact point does not move at all — it lies on the axis, so no translation occurs anywhere at the interface. Nothing about position changes. The only quantity that changes is the relative angle between the two surfaces at the point where they meet.

Spin at a contact is a phase difference and nothing else. No position, no distance, and no separation registers it.

That is the free circle from Section 4, appearing as a motion rather than as a residual symmetry. The mark fixed an axis and left the rotation about it undetermined; the twist is a body moving in exactly that undetermined direction. Its state is a point on the circle and its change is an advance along it, which is what phase is.

5.3 The free direction is the untransmittable direction

The two facts now available — that a contact leaves rotation about the normal free, and that spin about the normal is the motion in that direction — combine into a statement about what a contact can do.

An ideal frictionless contact transmits force only along the common normal. The position vector from a body's centre to the contact point is also along that normal. So the torque exerted on the body is $\mathbf{r} \times \mathbf{F}$ with both vectors parallel:

$$\mathbf{r} \parallel \mathbf{n} \text{ and } \mathbf{F} \parallel \mathbf{n} \Rightarrow \mathbf{r} \times \mathbf{F} = \mathbf{0} \text{ identically}$$

The torque about the contact normal is zero for any force magnitude whatsoever. An ideal contact exerts no twisting influence about the axis it defines, and cannot be made to by pressing harder.

The symmetry a contact leaves free is exactly the direction in which the contact can transmit nothing. These are the same one-dimensional space.

This is not a coincidence between two results. It is one result with a geometric face and a dynamical face. Section 4 computed the residual stabiliser by asking which rotations fix the marks; the calculation here computes the same subspace by asking which torques the contact cannot deliver. A mark leaves free precisely what the contact that made it cannot touch.

5.4 Abstract and material: the same change, two prices

The phase can now be described from both ends of the chain of Section 1, and the difference between the descriptions is a difference in cost rather than in content.

	Abstract \u2014 relation only	Material \u2014 dynamics
State	relative phase, a point on a circle	relative orientation of two bodies
Change	advance along the circle	angular acceleration
What it costs	nothing; it is a free coordinate	torque about the contact normal
Source of that cost	none available	spin friction, and nothing else
Closure	\u03b8 identified with \u03b8 + 2\u03c0	the same, plus stored elastic twist

In the abstract the change is the phase: a point moving on the free circle, costing nothing, because the circle is what the mark left behind and nothing constrains motion along it. In matter the same change requires angular acceleration, which requires torque, and the only torque available about that axis is spin friction.

So the consequence for an ideal contact is sharper than it first appears. Remove friction and the relative phase becomes free to be anything and impossible to alter. The freedom and the immovability are the same property, seen from the two sides: a coordinate that nothing constrains is a coordinate that nothing can drive.

This gives the price of a transformation an exact name in one case. A change of phase at a contact costs friction. Where friction is absent the price cannot be paid, and the transformation does not occur — not because it is forbidden but because there is no mechanism to purchase it. The coordinate remains, unconstrained and unreachable.

6. What distinction and continuation generate

Set the sphere aside. It was useful because it shows what happens once a metric exists; it says nothing about where the metric came from. This section starts from primitives that assume no metric at all.

Take a system of distinguishable states with lawful transitions between them. Distinguishability supplies the states; lawful transition supplies the arrows. Nothing further has been assumed — no distance, no coordinates, no continuity, no dimension. The object is a directed graph and that is all it is.

Define the cost of getting from one state to another as the fewest lawful steps required:

$$\kappa(x,y) = \min \{ n : \text{there is a path } x \rightarrow \dots \rightarrow y \text{ of length } n \}$$

This is the only definition available at this level, because counting steps is the only operation the structure supports. It costs nothing beyond the graph. Now test the metric axioms against it, one at a time, and see which are consequences and which are additions.

6.1 Four axioms are forced

Comparison. Path lengths are natural numbers and natural numbers are ordered, so a comparison between two transformations exists without further apparatus. Given x , one can ask whether reaching y costs more or less than reaching z , and the answer is determined. This is the weakest of the properties and the most easily overlooked, since it is usually taken for granted before any metric is mentioned.

Magnitude. The count itself is the magnitude, and it is an integer. This is worth emphasising because it is the first place the conventional apparatus overreaches. The primitives do not supply a real number. They supply a member of the natural numbers, and every real-valued quantity used later in geometry is a purchase whose price Section 6 states.

Identity. $\kappa(x,x) = 0$, because the empty path exists. And $\kappa(x,y) > 0$ for distinct x and y , because no path of length zero connects distinct states. Both directions of the identity axiom are immediate from the definition.

Triangle inequality. If there is a path of length $\kappa(x,y)$ from x to y , and a path of length $\kappa(y,z)$ from y to z , then concatenating them gives a path from x to z of length $\kappa(x,y) + \kappa(y,z)$. The minimum over all paths from x to z is therefore no greater than that sum:

$$\kappa(x,z) \leq \kappa(x,y) + \kappa(y,z)$$

This is a proof rather than an observation, and it makes clear where the axiom comes from: concatenation of paths is itself a path. The triangle inequality is the shadow of composition. Measured on a random directed graph over 40 states with edge probability 0.12, there were zero violations across all 64,000 ordered triples, which confirms the implementation and adds nothing to the proof.

6.2 Symmetry is not forced, and fails generically

Nothing in either primitive gives $\kappa(x,y) = \kappa(y,x)$. Distinguishability says nothing about reversal, and lawful continuation from x to y is a different fact from lawful continuation from y to x . The definition of κ does not impose it, and no argument from concatenation produces it.

On the measured graph, 504 of 780 pairs reachable in both directions had unequal costs — 64.6 percent. Asymmetry is not a corner case. It is the generic situation, and the symmetric cases are the exceptions. Explicit instances from that run:

Pair	$\kappa(x,y)$	$\kappa(y,x)$
(0, 1)	2	1
(0, 2)	3	2
(0, 4)	3	2
(0, 7)	3	2
(0, 8)	3	1

In the last of these, reaching state 8 from state 0 requires three transitions while returning requires one. There is nothing pathological about the graph that produces this; it is what a directed structure does when nothing constrains it to be otherwise.

6.3 The result is a known object

What these primitives generate is a quasi-metric: identity and the triangle inequality without symmetry. This is not a shortfall or an intermediate stage on the way to something better. It is a standard mathematical object with a precise characterisation, identified half a century ago.

Lawvere showed in 1973 that a metric space is exactly a category enriched over the non-negative extended reals, with the order reversed and addition as the tensor. Under that identification the enrichment axioms translate directly:

Enrichment axiom	Metric translation
identity morphism at each object	$d(x,x) = 0$
composition of morphisms	$d(x,z) \leq d(x,y) + d(y,z)$
no corresponding axiom	$d(x,y) = d(y,x)$

The enrichment gives identity and composition and nothing else. Symmetry has no counterpart among the axioms, and Lawvere argued explicitly that it is not categorically natural — that requiring it is an extra condition imposed for familiarity rather than a structural necessity. Objects and morphisms give a quasi-metric, exactly.

So the arrival at a quasi-metric here is not a failure to reach a metric. It is arrival at the correct object, and the earlier search for a derivation of symmetry was a search for something that should not have been expected.

6.4 Symmetry is forbidden, not merely underived

The stronger claim is that symmetry contradicts a constraint the framework needs elsewhere, and this section states the argument rather than asserting the conclusion.

A transformation carrying x to y is not the same transformation as one carrying y to x . They have different sources, different targets, and different effects on any structure that tracks where a state came from. Reversal

produces a second transformation with opposite orientation; it does not undo the first. If it did, the composite would be indistinguishable from having done nothing, and the record that anything occurred would be gone.

That record is what makes transformation distinguishable from stasis. A system in which every transition is annihilated by its reverse has no way to witness that a transition happened, because the witness is precisely the difference between the state before and the state after, and running both directions returns the original state with no residue.

A symmetric distance quotients direction. It maps the ordered pair (x, y) and the ordered pair (y, x) to a single number, discarding the ordering. So imposing $d(x, y) = d(y, x)$ discards exactly the quantity that distinguishes a transformation from its reverse.

The geometry generated by distinction and lawful continuation is directed. Symmetry is purchased with reversibility, which is a condition on transitions rather than a property of distinctions.

Reversibility is a substantive assumption and it is not universally available. It says that every transition can be undone at equal cost, which is a strong claim about the transition structure and a false one in most systems anyone actually studies. It happens to hold in the settings where Riemannian geometry applies, which is why the symmetry axiom has looked like a triviality rather than a purchase.

7. The price list

With symmetry correctly identified as a purchase, the remaining structure prices out cleanly. Two rungs are free and everything above them is bought.

7.1 The integers and the rationals are free

Counting steps gives the natural numbers directly. The count exists because paths have lengths and lengths are counts; no axiom is added and no structure imported.

Comparing two counts gives their ratio, and a ratio of integers is rational by construction. So the rationals arrive as soon as two transformations can be compared, which Section 6.1 established is available immediately. Neither rung costs anything.

This is a stronger statement than it may appear. It says the number system that a transformation structure natively supports is the rationals, and that everything requiring more than the rationals is an addition to the structure rather than a discovery within it.

7.2 The reals cost two separate things

The passage from the rationals to the continuum is normally charged as a single step. It is two, and they are independent.

The first is a scale. Consider the cycle graph on n vertices with its path metric, and compare it to the circle. Normalising each edge to length $2\pi/n$, the discrete metric agrees with the circle's geodesic metric to a maximum deviation of 8.9×10^{-16} at every size tested from 8 vertices to 8192 — exact convergence, limited only by floating-point representation.

But the normalisation was supplied from outside. The graph gives counts. Nothing in it names an edge length, and nothing in it prefers $2\pi/n$ over any other choice. Choosing one is a mark, and it is the same kind of purchase

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

as choosing an origin on a sphere: a distinguished element of a set that contains no distinguished element. The convergence is real and the scale is not free.

The second is completeness. Every Cauchy sequence of rationals converging to a rational limit is a closure condition, and no discrete structure possesses it. It is an axiom, it is independent of the scale, and it is what actually separates the rationals from the reals.

Rung	What it costs	What it buys
Distinguishable states	C_0	the objects
Directed continuation	C_1	a directed graph
Path count	concatenation	the integers \(\mathbb{Z}\) free
Ratio of counts	comparison	the rationals \(\mathbb{Q}\) free
Quasi-metric	identity and triangle	forced; four axioms of five
Metric	REVERSIBILITY	symmetry
A length	A MARK	the edge scale, the normalisation
Continuum	COMPLETENESS	the reals, and the irrationals with them
Unbounded addressing	an irrational ratio	density, by Weyl equidistribution

7.3 What completeness is for

The last rung is not decoration and the reason to buy completeness is concrete.

Consider a staircase generated by an irrational slope — the sequence of increments produced by taking successive integer multiples of a real number and recording how the integer part jumps. If the slope is rational with denominator q , the sequence is periodic with period q , and there are exactly q distinct phases. If the slope is irrational, the sequence is aperiodic and the orbit of its fractional parts is dense in the unit interval, by Weyl's equidistribution theorem.

So the number of distinguishable addresses generated by such a staircase is finite in the rational case and unbounded in the irrational one. Any addressing scheme resting on an irrational ratio is therefore resting on completeness, whether or not it declares this, and the density that makes the addressing non-degenerate is exactly what the axiom purchased.

8. What is recoverable without an exterior

There is an obvious objection to a framework in which no exterior vantage returns a value, and it should be met directly. Geometry plainly has numbers. Curvature is a quantity, radius is a quantity, and if nothing stands outside a surface then something must be supplying them from somewhere. The objection has an answer and the answer is a theorem.

Gauss's Theorema Egregium, 1827, establishes that Gaussian curvature is intrinsic: it is determined entirely by measurements made within the surface, with no reference to any embedding. A surface does not need to be sitting inside a larger space for its curvature to be defined, and no exterior observation is required to obtain it.

This is not an analogy imported for convenience. It is the theorem the framework needs, proved two centuries ago for other reasons.

8.1 Two independent constructions, one invariant

Two operational routes recover the radius of a sphere, and neither is supplied the answer. That they agree is the content of intrinsic curvature made concrete.

The local route uses two equal geodesic arcs at right angles and their diagonal. On a sphere the exact relation for a right spherical triangle is

$$\cos(c/R) = \cos(a/R) * \cos(b/R)$$

verified to twelve decimal places at every scale tested, including the degenerate quarter-turn case where all three arcs equal $\pi R/2$. Expanding for small arcs gives

$$c^2 = a^2 + b^2 - a^2 b^2 / (3 R^2) + \dots$$

The flat Pythagorean relation is the leading term and the correction carries the curvature. So the flat relation is not the primitive geometry from which the spherical one is derived; it is the limit of the spherical relation as the radius grows without bound. Solving the deviation for R recovers 0.999833 from arcs of length 0.05 and converges to unity as the arcs shrink.

The global route transports a frame around a closed loop and measures its failure to return unchanged. On a sphere the holonomy angle equals the enclosed area divided by R^2 , and around a latitude circle at angle ϕ it is $2\pi(1 - \sin \phi)$, matching the enclosed cap area to six decimal places at every latitude from the equator to the pole. This is Gauss-Bonnet, and solving it for R recovers the radius as well.

Run against spheres of radius 0.5, 1, 2, 5, and 10, the two routes agree: 0.499917 against 0.500000, 0.999833 against 1.000000, and so on to 9.998334 against 10.000000. The residual error is the truncation in the local expansion and shrinks with arc length.

The agreement is not a coincidence and it is not a proof of the framework. Both constructions compute aspects of the same intrinsic quantity, and Theorema Egregium is why they must be measuring the same thing. What the agreement demonstrates operationally is that neither construction consulted an exterior, which is the point at issue.

8.2 A weaker starting point still suffices

Both routes above use spherical trigonometry, which assumes the geometry is spherical. A stricter experiment supplies less and still recovers the geometry.

Give an algorithm only a matrix of pairwise distances. No coordinates, no embedding, no geometry named, no curvature supplied. Classical multidimensional scaling forms the Gram matrix from the squared distances and examines its eigenvalues: a Euclidean distance matrix produces a positive semidefinite Gram matrix, and a non-Euclidean one does not.

Applied to 60 points, flat points in three-space gave a most-negative eigenvalue of -4.3×10^{-14} — numerical zero, Euclidean. Geodesic distances on a unit sphere gave -8.89 , and on a sphere of radius 5 gave -218 . The matrix alone reveals that the space is not flat.

Fitting the radius uses the chord map. On a sphere of radius r , a geodesic of length d subtends a chord of length $2r \sin(d/2r)$, and chords are Euclidean only when r is correct. Sweeping candidate radii and locating where the Gram defect vanishes recovers 0.998664 from a true radius of 1, 2.995993 from 3, and 6.990651 from 7, with the defect at the true radius at the level of 10^{-13} and rising sharply on either side.

What remains assumed in that experiment is exactly one thing, and naming it precisely is the point of running it: a symmetric real-valued relation obeying the triangle inequality. Distinguishability and lawful continuation give neither the symmetry nor the real values, as Sections 6 and 7 established. The distance matrix is the purchase; everything geometric follows from it and nothing geometric precedes it.

8.3 The criterion: invariance decides

Sections 8.1 and 8.2 show that curvature needs no exterior. Section 2.1 shows that no point of a sphere is distinguishable from any other, so position cannot be recovered at all without a mark. These look like they are in tension and they are not, because they concern different kinds of quantity, and the line between them is exact.

Apply the group to an entire configuration and ask which quantities come back unchanged. Over 2000 random rotations applied to a sphere of radius 2.3 carrying two marked points:

Quantity	Spread under the group	Status
Radius	2.7×10^{-16}	invariant
Gaussian curvature	5.4×10^{-17}	invariant
Total curvature	5.4×10^{-15}	invariant
Geodesic separation of the two marks	6.9×10^{-16}	invariant
x-coordinate of a point	1.27	NOT invariant
Longitude of a point	1.89	NOT invariant

A quantity is recoverable without a reader exactly when it is invariant under the symmetry group. That is the entire dividing line.

Invariant quantities are properties of the whole — radius, curvature, total curvature, and the separation of points that have already been marked. Non-invariant quantities are properties of a single part — position, orientation, longitude, which point. The first kind is intrinsic and needs nothing outside. The second kind requires a mark and cannot be had without one.

So Theorema Egregium gives intrinsic curvature and does not give intrinsic position, and nothing could give the latter, because points on an orbit are not distinguishable. Stated together: the surface knows its own shape, and it does not know where you are on it.

The consequence is worth making concrete. On a perfectly featureless planet, with no stars, no landmarks, and no external reference of any kind, a walker can recover the curvature of the world by the local construction of Section 8.1 — two equal arcs at right angles and their diagonal. What the walker cannot recover is longitude,

because curvature is invariant and longitude is not. One would know the size of the world and never know where one stood on it.

9. Matter, and the two kinds of state

The outermost closure is not a material object. It is the closure of the address structure — the boundary past which no further coordinate, value, or reference is available within the model.

The weighting operation does not apply to it, and the reason is a matter of domain rather than of force. Weight is the response to a gradient, a gradient requires two distinguishable states to be evaluated between, and the model supplies no second state beyond the outermost boundary. So the operation has no second argument.

The mathematical closure has no external weighting relation.

This is a claim about the domain of definition of an operation within this framework and it should not be read as a claim about the gravitation of a physical object. A physical sphere has weight because there is something outside it to weigh it against. A mathematical closure is not a test body and has no exterior term. The difference between those two situations is the whole of the point, and collapsing them would be the same category error the paper spends Section 1 forbidding.

Inside the closure, gradients exist, because the enclosed domain contains distinguishable states and transitions between them. And weight is inherited downward from enclosure in exactly the way address is. A body's weight at a given radius depends only on what is enclosed beneath it; every shell it lies inside contributes precisely nothing. For a uniform ball, the field at radius r is proportional to r within the body and falls as $1/r^2$ outside it, and the mass at larger radii is invisible to the calculation. That is the shell theorem, and it is the physical form of the interior-supplies-nothing statement.

So there are two kinds of state and they should not be merged. The mathematical closure determines what address structure is available. The material domain is what occupies the relational structure that closure generates. These are not two regions with one inside the other; they are two readings, in the sense of Section 3.2. The apparent outside of the whole is not another material region awaiting description. It is the complementary reading of the same closure.

How the first generates the second is not established here. That the closure supplies the address structure within which matter becomes addressable is a framework position, and the derivation of material structure from relational closure is the largest open item this paper leaves. Section 13 states it as such.

10. Consequences

Three consequences follow from the preceding sections and each contradicts a standard assumption. They are stated together because they are the same consequence at three scales.

10.1 Riemannian geometry is a sector, not a foundation

The familiar apparatus — symmetric distance, smooth manifolds, a metric tensor, curvature as a tensor field — is not what distinguishability and lawful continuation produce. It is what they produce after three purchases: reversibility, a scale, and completeness. Each is nameable, each is independent of the others, and none is free.

So the symmetric world is a sector of a directed transformation structure, selected by the condition that every transition can be undone at equal cost. That condition is substantive rather than trivial. Where it fails, the geometry is asymmetric and quasi-metric, the machinery of Riemannian geometry does not apply, and there is no reason internal to the primitives why the condition should hold everywhere.

This is not a claim that Riemannian geometry is wrong. It is a claim about its position: it describes a sector characterised by an assumption, and the assumption has been invisible because the sectors we work in satisfy it.

10.2 Space is a reading, not a container

The conventional order of dependency runs space, then distance, then paths, then transformations. The order established here runs the other way: transformations, then paths, then path cost, then geometry. Space is not the arena in which transformations occur. It is the representation generated by the relations among them.

Section 2 is the cleanest instance. The sphere required no ambient space to be defined — an orbit needs a group and a point, and the three-dimensional container in which we habitually draw it is a later convenience the definition never used. Section 3.1 sharpens this: the whole of three-space, read through the rotation group, is a ray.

10.3 Magnitude begins as a count

The primitive quantity is not a real-valued measurement of a pre-existing continuous extent. It is the number of lawful transformations traversed, and it is an integer. Continuous mathematics is a later representation of a discrete transition structure, reached by paying for a scale and a completion.

An integer index is therefore not a number describing reality from outside it. It is an address along a transformation history, and the real-valued geometry that eventually appears is a representation of that history rather than the ground the history stands on. This reverses the usual reading, in which discreteness is an approximation to an underlying continuum; here the continuum is a construction on top of the discrete, and the construction has a price.

11. Relation to existing work

Almost every mathematical result used in this paper is established and none of the classical ones is claimed as new. Stating precisely which is which is necessary, because the paper's contribution is an arrangement rather than a discovery, and an arrangement is only worth something if the reader can see exactly what was rearranged.

Result	Source	Used for
Orbit-stabiliser relation	standard group theory	dimension of the sphere from the group alone (\u00a72.3)
Rodrigues rotation formula	Rodrigues, 1840	explicit transitivity of $SO(3)$ on the orbit (\u00a72.1)

Result	Source	Used for
Jordan-Brouwer separation	Brouwer, 1911	exactly two regions from closure, one bounded (\u00a73.2)
Shell theorem	Newton, 1687	field identically zero inside a uniform shell (\u00a72.2)
Divergence theorem / Gauss's law	Gauss, 1813	the field is a surface integral (\u00a72.2)
Theorema Egregium	Gauss, 1827	curvature intrinsic; no exterior needed (\u00a77)
Gauss-Bonnet	Gauss / Bonnet, 1848	holonomy equals enclosed curvature (\u00a77.1)
Metric space as enriched category	Lawvere, 1973	symmetry is not an enrichment axiom (\u00a75.3)
Equidistribution theorem	Weyl, 1916	irrational slope gives dense addressing (\u00a76.3)
Classical multidimensional scaling	Torgerson, Gower	curvature from a distance matrix (\u00a77.2)

11.1 What is not claimed

It is not claimed that spherical trigonometry differs from flat trigonometry, that parallel transport on a curved surface produces holonomy, that Gaussian curvature is intrinsic, or that a metric space is a special kind of enriched category. All four are known, and three of the four are more than a century old.

It is not claimed that any calculation in this paper is beyond a first course in differential geometry or group theory. The computations are elementary by design; the argument depends on their arrangement and not on their difficulty.

It is not claimed that Riemannian geometry is incorrect or that its results fail. The claim is about its position in a dependency order, and Section 9.1 states it in the narrowest form that carries the point.

11.2 What is claimed

The arrangement. Taking the orbit as the definition of the sphere rather than as a property of it, and deriving the absence of a privileged point, the absence of a gradient at the centre, the dimension, and the single-datum structure of the nested family from that one starting point.

The price list. Identifying reversibility, a scale, and completeness as three separate purchases separating the primitives from Riemannian geometry, and locating exactly which metric axioms each one buys. The individual observations are not new; the accounting is.

The reading of Lawvere's result as an argument that symmetry should not be expected rather than as a technical characterisation. Lawvere made the mathematical point; the use of it to explain why a search for a derivation of symmetry was misdirected is the addition here.

The identification of the residual symmetry after a contact with the direction in which that contact transmits no torque. The stabiliser calculation and the vanishing of $r \times F$ for parallel vectors are each elementary; that they compute the same one-dimensional subspace, and that this is why an ideal contact cannot alter a relative phase, is the addition here.

The two-readings formulation of a closure, and the consequence that nesting folds rather than terminating. Section 3.2 is the part of the paper least supported by prior literature and most in need of the scrutiny it will receive.

12. Claims and their status

Every substantive claim in the paper appears below with the basis on which it rests. The three categories are: theorem, meaning established mathematics used as stated; derived, meaning a consequence of the paper's own definitions; and framework, meaning a position taken within this account that is coherent but not forced by the mathematics alone. Numerical entries note the verification performed.

§	Claim	Basis
1	Shape determines admissible relations; values are last	framework ordering
1.1	Logic invariant; mathematics its description; shape its manifestation	framework
2	The sphere is the orbit of a point under $SO(3)$	definition
2.1	$SO(3)$ acts transitively on the orbit	theorem; Rodrigues construction, 500/500 checked
2.1	No point of a sphere is distinguishable from any other	derived from transitivity
2.2	The origin is the unique fixed point of $SO(3)$ on R^3	theorem; proof given
2.2	No gradient is definable at the fixed point	derived; transitivity on directions
2.2	The fixed point admits no weighting relation	derived; a domain-of-definition claim
2.2	Field identically zero inside a uniform shell	Newton; verified numerically
2.3	S^{n-1} is the orbit of a point under $SO(n)$	definition + orbit-stabiliser
3.1	$R^3/SO(3)$ is the half-line	theorem; proof given
3.1	Rotation-respecting structure is a function on a ray	derived
3.2	A closed surface forces exactly two regions, one bounded	Jordan-Brouwer
3.2	Bounded/unbounded and normal orientation are distinct asymmetries	derived
3.2	A boundary is one closure under two orientations	framework
3.2	Nesting folds rather than terminating	framework
4	Marks consume symmetry; the ledger saturates at three	theorem; rank computation
4	One mark gives an axis, not an angle	derived
4	An antipodal second mark costs nothing	derived; stabiliser stays dimension 1
4	The third mark resolves a discrete ambiguity	derived; reflection has determinant -1
5.1	κ satisfies identity, positivity, triangle	proved from concatenation
6.1	The primitive magnitude is an integer	derived
6.2	Symmetry fails generically	measured: 504/780 pairs, 64.6%

§	Claim	Basis
6.3	Metric space = category enriched over $[0, \infty]$	Lawvere 1973
6.4	A symmetric distance quotients direction	derived
6.4	Symmetry is bought with reversibility	derived
7.1	N and Q are free	derived
7.2	R costs a scale and completeness, separately	derived; cycle→circle to 10^{-16}
7.3	Irrational slope gives dense, unbounded addressing	Weyl
8	Curvature is intrinsic	Theorema Egregium
8.1	Two independent routes recover the same radius	verified, $R = 0.5$ to 10
8.2	Curvature recoverable from a distance matrix alone	verified to 3 decimals
9	Weight is inherited downward from enclosure	shell theorem
9	The mathematical closure has no external weighting relation	framework; domain of definition
10.1	Riemannian geometry is the reversible, scaled, complete sector	derived

13. What must follow

Four obligations are forced by the argument and not discharged here. They are targets with definite shape rather than disclaimers.

The characterisation of reversible sectors. Reversibility buys symmetry, but nothing in this paper says which transition structures possess it or what enforces it where it holds. Every symmetric result in the second half, and the entire literature the first half draws on, rests on that condition. A characterisation of the structures in which it holds is the first requirement, and it is well posed: given a directed transition structure, when is the induced quasi-metric symmetric?

The origin of the scale. The normalisation carrying a discrete path count into a continuous length is a mark, and this paper treats it as a purchase without asking what fixes its value. Whether any transformation structure selects a preferred scale, or whether the choice is genuinely free, is open and consequential: if it is free, then every metric in physics carries a parameter that was never derived.

The general boundary theorem. Every geometric quantity used here is rewritable as a boundary quantity, and the divergence theorem supplies this wherever the integrand admits. A general proof that no interior quantity exists — that every quantity of a body is expressible at its boundary — is the principal outstanding debt of this account and it is a well-posed problem.

The asymmetric geometry itself. If the primitive structure is directed, the natural geometry is quasi-metric and Finsler-like rather than Riemannian, and the physics of such a structure is not developed here. That is where the remaining content of this account lies, and it is the reason the price list matters: it says what would have to be given up.

14. Summary

A sphere is what one point does under rotation. It has no privileged element because an orbit has none, no address until a mark is made, and no gradient at its centre because the centre is the group's fixed point and the group makes every direction there equivalent. Its interior is not a region it contains but a family of other orbits, and the quotient of three-space by the rotation group is a ray, so the whole family carries a single datum.

A mark buys structure by consuming symmetry, and the symmetry is finite. Three marks exhaust it — two continuous dimensions, then one, then a discrete bit of handedness — and a fourth adds content within a frame rather than more frame. A mark placed where structure already exists is free and produces nothing.

Distinguishability and lawful continuation give a directed graph. Path count on that graph gives identity, positivity, and the triangle inequality, forced by concatenation. It does not give symmetry, and it should not: reversal is a second transformation rather than an undoing, and a symmetric distance discards exactly the direction that distinguishes transformation from stasis. Lawvere's characterisation says the same thing from the categorical side, and symmetry is not among the axioms.

What is normally called geometry is that structure after three purchases. Reversibility buys symmetry. A mark buys a scale. Completeness buys the continuum, and with it the irrationals that make addressing unbounded. The integers were free and the rationals were free and nothing above them was.

Space is not the container in which this happens. It is the representation of the relations among the transformations, and the sphere — the cleanest object in it — turns out to be one point, moved.

The orbit is the mathematical closure. What becomes addressable within its relational domain is the material world, and how the first generates the second is the question this paper leaves open and has earned.